1st: Aspects 10 & 11 (Make sense of and switch between different representations)

- **Analysis:** This is the most critical part of the problem. A student must integrate two different 2D representations—the "Front view" and the "Side view"—to construct a mental mathematical model of the 3D roof. They must identify the width of one roof section (2.50m) from the front view and the length (6.00m) from the side view. This ability to switch between and relate different views is a prerequisite for solving the problem (p. 1).

2nd: Aspects 6 & 7 (Extracting and manipulating mathematical information from diagrams)

- **Analysis:** The problem requires extracting key values from plans that are "not to scale." The student must purposefully identify which numbers are necessary for calculating the area of the rectangular roof sections (2.50 and 6.00) while ignoring distractor data (such as the 2.40m wall height or 2.00m door width). This involves filtering and manipulating geometric information (p. 1).

3rd: Aspect 8 (Articulate a solution, showing intermediate mathematical results)

- **Analysis:** The prompt explicitly states, "Show your work." This means the student cannot simply provide a final answer; they must clearly present intermediate steps (e.g., calculating the area of one side first: $2.5 \times 6 = 15$). This reflects the ability to summarize and present mathematical procedures (p. 1).

4th: Aspect 12 (Use a multi-step procedure leading to a solution)

- **Analysis:** Solving this requires a clear logical sequence:
  1. Recognize the roof consists of two identical rectangular sections.
  2. Calculate the area of one rectangle (2.5×6).
  3. Multiply that result by 2 to find the total.
     This is a standard multi-step procedure leading to a mathematical conclusion (p. 1).

5th: Aspects 1 & 5 (Perform a simple calculation & utilise familiar algorithms)

- **Analysis:** The final act of multiplication and the use of the rectangle area formula ($A = l \times w$) are applications of basic mathematical rules and simple calculations.

第一位：方面 10 & 11 (Make sense of and switch between different representations)

- **分析：** 这是本题最具挑战性的部分。学生必须整合“正面图（Front view）”和“侧面图（Side view）”这两个不同的 2D 表达方式，来构建屋顶这一 3D 对象的数学模型。他们需要从正面图中确定屋顶单侧的斜边长度（2.50m），再从侧面图中确定屋顶的长度（6.00m）。这种在不同视图间切换并建立关联的能力是解题的前提。（p. 1）

第二位：方面 6 & 7 (Extracting and manipulating mathematical information from diagrams)

- **分析：** 题目要求从非比例绘制的平面图中提取关键数值。学生必须有目的地识别出哪些数字是计算“矩形屋顶面积”所必需的（2.50 和 6.00），并忽略干扰项（如墙高 2.40 或门宽 2.00）。这涉及到对几何图形信息的筛选、提取与操作。（p. 1）

第三位：方面 8 (Articulate a solution, showing intermediate mathematical results)

- **分析：** 题目明确要求“Show your work（写出计算过程）”。这意味着学生不能只给出一个最终结果，必须清晰地展示中间步骤（例如：单侧屋顶面积 = $2.5 \times 6 = 15$），这反映了对数学解题过程的表达与总结能力。（p. 1）

第四位：方面 12 (Use a multi-step procedure leading to a solution)

- **分析：** 解决此题需要一个清晰的逻辑步骤：
  1. 确定屋顶由两个相同的矩形组成。
  2. 计算其中一个矩形的面积（2.5×6）。
  3. 将该面积乘以 2 得到总面积。
     这属于一个标准的、指向最终答案的多步骤程序。（p. 1）

第五位：方面 1 & 5 (Perform a simple calculation & utilise familiar algorithms)

- **分析：** 最终的计算行为（乘法运算）和使用的公式（矩形面积 $A = l \times w$）属于基础数学规则和简单计算的应用。